

Determining what Factors are most associated with Pedestrian Injury Severity in Traffic Collisions

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1. Introduction

Examining the Impact of Land Use, Transit Stops, Roadway, and Demographic Factors on Pedestrian Injury Severity in Connecticut is the title of a dataset collected by the New England University Transportation Center for the purposes of examining what factors were most associated with pedestrian injury severity in traffic collisions from 2016 to 2025. Our analysis is specifically concerned with the factors most associated with fatal/severe pedestrian injury and no injury (suspected minor injuries from the original data set were removed). This data organizes the predictors into four broad categories: pedestrian, crash, vehicle, and driver, with various miscellaneous predictors also included. The pedestrian predictors describe a series of characteristics of the pedestrian involved in the crash, including location, actions, gender, etc. The driver predictors describe a series of similar characteristics for the driver of the vehicle involved in the crash. The crash predictors describe details related to the crash, such as roadway conditions. The vehicle predictors describe the vehicle involved in the crash. The goal of this statistical analysis is to determine which of these predictors are most associated with pedestrian injury status and what factors lead to more or less severe pedestrian injury in traffic collisions.

2. Methods

2.1 Data Tidying

The data used in this analysis records crash data in Connecticut from 2016 - 2025. In total, 14,645 crashes were recorded. However, because this study only examines crashes with severe/fatal or no injury outcomes, a large number of these observations were not considered. Furthermore, a series of predictors which weren't clearly defined, weren't of interest, or contained too many null values were removed. Consult the appendix to see which predictors were specifically removed.

The remaining predictors were modified to better fit the models employed in this analysis: a classification tree model, a random forest classification model, and a series of logistic regression models. The majority of predictors in this dataset are categorical with an unknown category to capture missing values. All null observations from the raw data were merged into the unknown category of the corresponding predictor variable. Other categories with vague meanings were also collapsed into the unknown category of many predictors. Consult the appendix to see specific changes.

Because interpretability was a priority in this analysis, a number of predictors' categories were collapsed into simpler, more readable categories. For example, the *PEDESTRIAN_Actions.Prior.to.Crash* predictor was collapsed down to three categories: *Crossing*, *Moving Along Roadway*, and *Working in Trafficway*. Given the complexity of this dataset, the goal of these mutations was to simplify the predictors so the results of our models, specifically the tree-based models, were more interpretable.

One of the biggest changes made to the data set were the changes to the response variable of interest *PEDESTRIAN_Injury.Status*. This variable originally contained five categories: *Fatal_Injury (K)*, *Suspected Serious Injury(A)*, *Suspected Minor Injury (B)*, *Possible Injury (C)*, *No Apparent Injury (O)*. Observations corresponding to the categories B and C were filtered out of the dataset, and categories K and A were combined. The result was a binary response variable with categories *Suspected Serious/Fatal injury (A/K)* and *No Apparent Injury (O)*, with a roughly even split between them. This choice was made to improve the interpretability of our results by focusing on a narrower subset of the whole dataset. With these changes made, our results would help us determine which of the predictors are most associated with fatal/severe injuries vs. no injuries at all, as mentioned previously. Finally, any remaining observations with null values were removed. In the end, this left 4606 observations remaining.

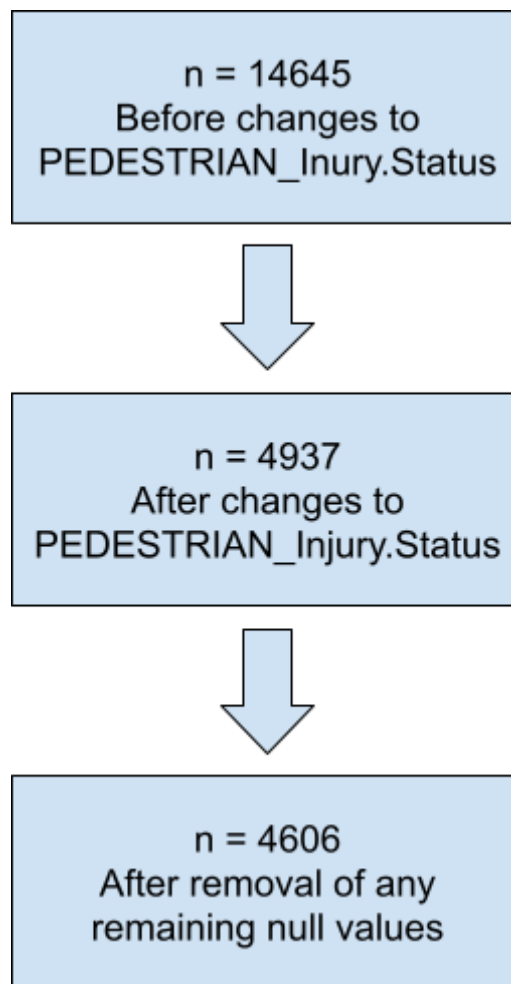


Figure 1: Consort diagram illustrating change in n after filtering.

2.2 Statistical Analysis Plan (Switch to past tense)

The first model we consider is a simple classification decision tree. The goal of this model was to find the predictors most associated with the outcome, *PEDESTRIAN_Injury.Status*. The benefit of this model was that it was easily interpretable and it provided us with a couple of predictors to focus on. The data was split into training and test sets. 70% of the recorded observations were allocated to the training set, and the remaining 30% were allocated to the test set. A contingency table and two decision trees, one from before pruning was applied and one from pruning was applied, were used to interpret the model. This test and training split was maintained across all three models.

The second model we fit was a random forest classification. The goal of this model was to verify that the predictors identified by the classification model were in fact significant and to identify other significant predictors missed by the initial classification tree. The two metrics pulled from the random forest model that we used to determine predictor importance were mean decrease in model accuracy, when a given predictor was removed, and mean decrease in the gini index, when a given predictor was removed. Predictors were selected from these importance metrics based on their performance and how interesting the interpretation of their association with the results were. The predictors of interest found by the classification and random forest model were analyzed further using the following model.

A logistic regression model was also employed to specifically compare the predictors identified by the previous two models. Individual predictors were fit to the model and likelihood ratio tests were used to test their significance in relation to the response variable.

3 Results

3.1 Classification Tree Results

When our data was fit to the classification tree model, it resulted in the two trees you see below. The tree on the right is the one which resulted after pruning was applied.

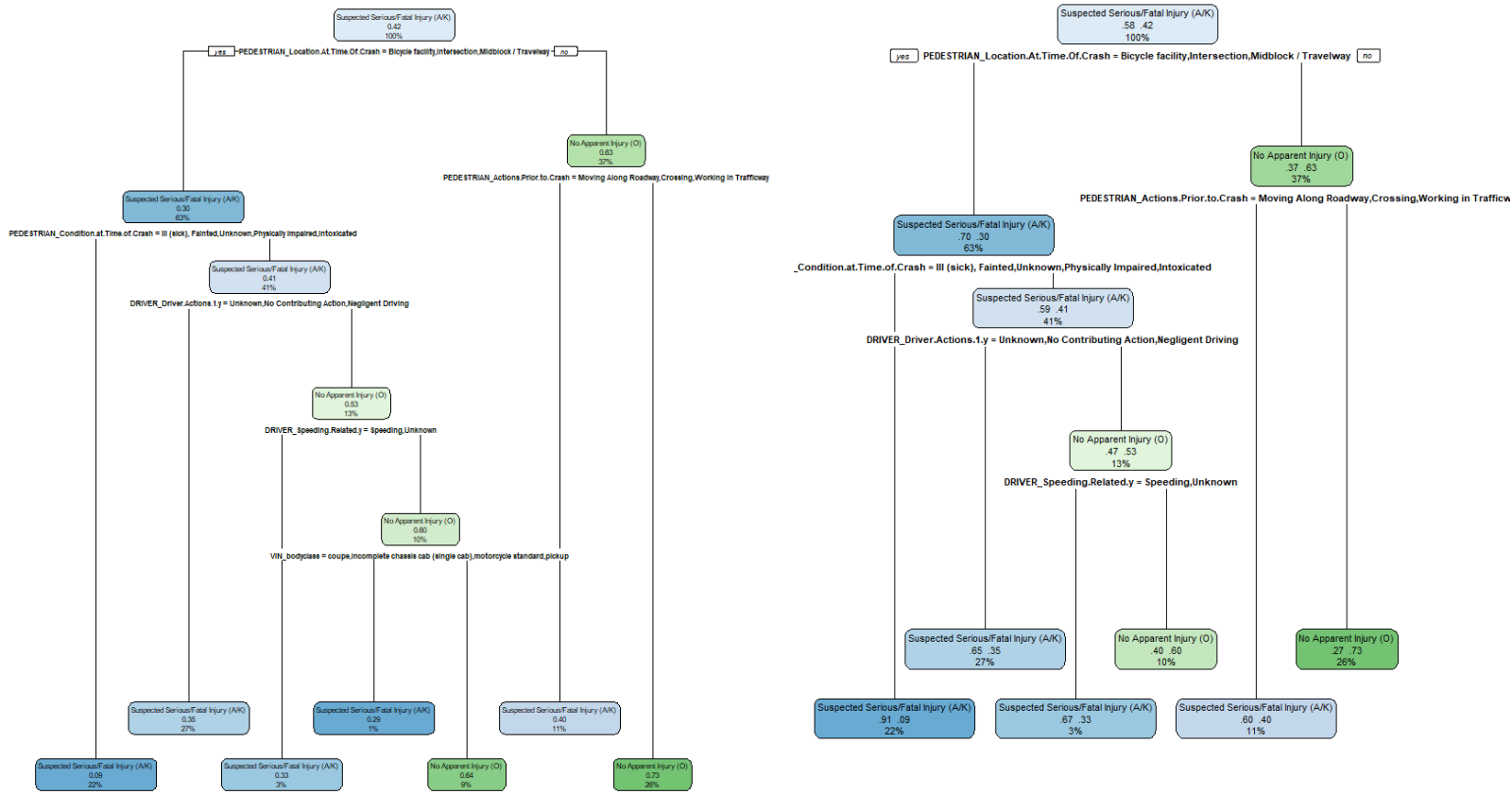


Figure 2: Classification decision tree before and after pruning. The tree on the left is before pruning, and the one on the right is after pruning.

As is illustrated by the pruned decision tree in *figure 2*, four variables of interest closely related to the outcome were pedestrian location, pedestrian actions prior to the crash, pedestrian condition at the time of the crash, and driver speed. According to this decision tree, pedestrian location was the strongest predictor of pedestrian injury severity, with bicycle lanes, intersections, and midblocks posing the highest risk of severe/fatal injury to pedestrians. Even if pedestrians are not in one of these locations, the decision tree reveals that if they are crossing, working, or moving along the road, they are still at higher risk of a serious/fatal injury.

3.2 Random Forest Regression Results

The random forest model, as mentioned previously, was fit to validate the importance of the variables identified by the decision tree and to identify additional significant predictors that the decision tree may have missed. Furthermore, we wanted to compare the performance of the two models, with the expectation that the random forest model would produce more accurate predictions on the test data. Variable importance and model performance is summarized below.

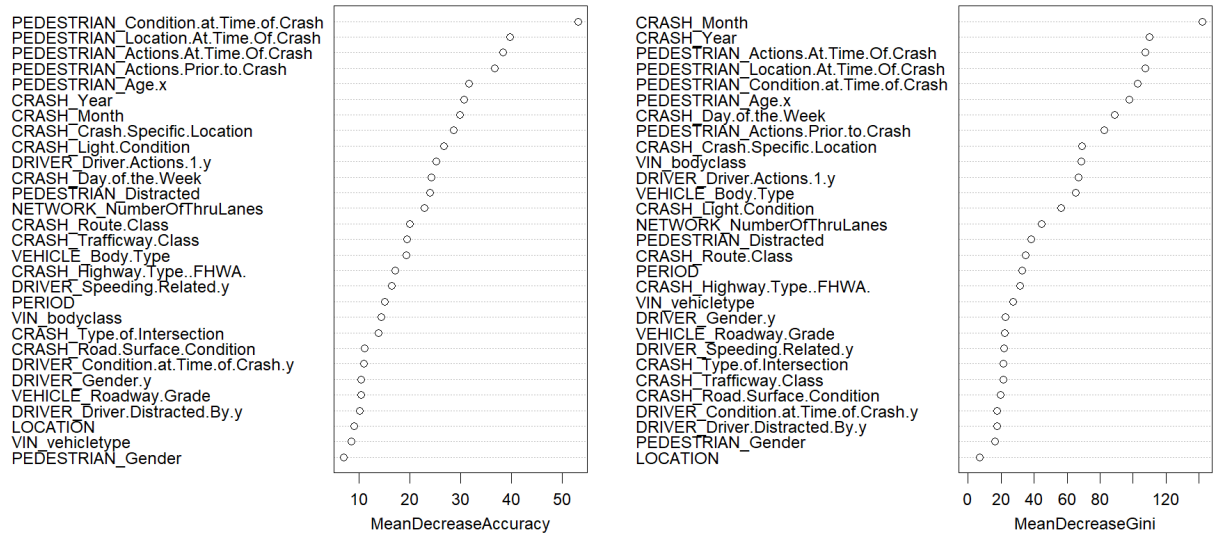


Figure 3: Importance of predictors generated from a random forest classification model.

	Classification Tree		Random Forest Classification	
	(A/K)	(O)	(A/K)	(O)
Suspected Serious/Fatal Injury (A/K)	657	245	687	163
No Apparent Injury (O)	140	262	110	344

Figure 4: Contingency tables for the classification tree and random forest classification models.

As is shown in figure 4, the random forest model generated better predictions than the classification tree, as was expected. In figure 3, all of the variables identified by the classification tree are shown to have high levels of importance, validating those earlier results. However, two new predictors, *CRASH_month* and *CRASH_year*; which did not appear in our classification tree results, are prominent in this output. To explore these results further, we employed isolated logistic regression models and an additional decision tree.

3.3 Logistic Regression Results

The *CRASH_Year* and *CRASH_Month* predictors were fit to respective logistic regression models, and likelihood ratio tests with a null model were run to verify their association with the response. P-values below 0.05 were recorded for both tests, meaning both predictors are associated with the response variable. To explore the crash year predictor further, a logistic regression with the predictors crash year and pedestrian actions prior to crash was fit as well as a

corresponding interaction model. A likelihood ratio test was run on these two models and it resulted in a p-value below 0.05, suggesting that there could be a relationship between year and pedestrian actions prior to crash. Finally, a decision tree was fit to just the *CRASH_Year* predictor, resulting in the figure below.

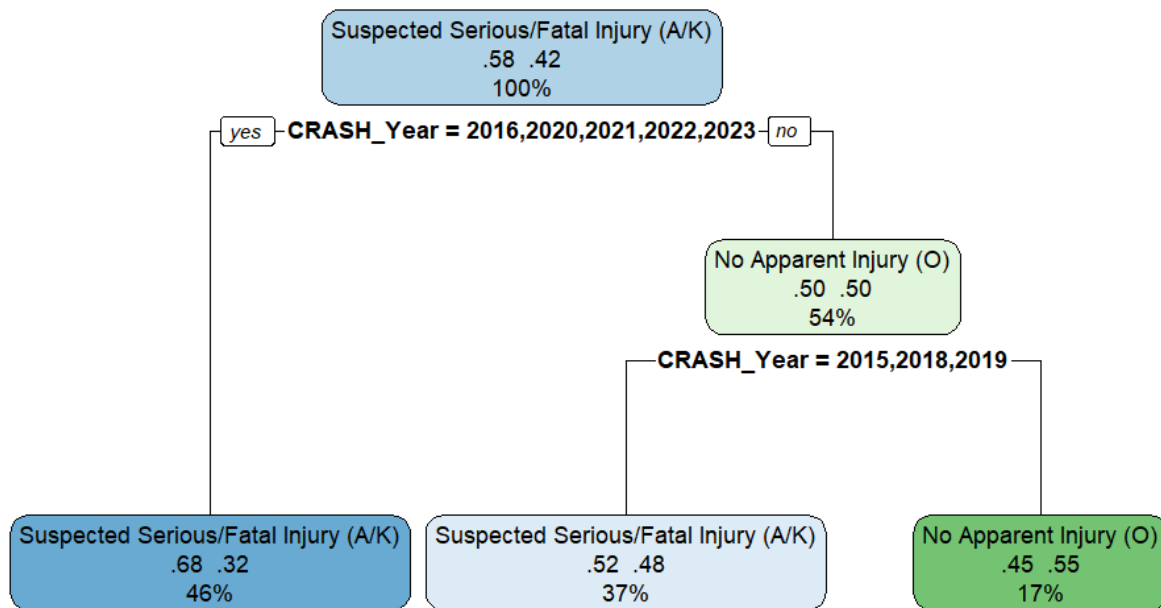


Figure 5: Decision tree illustrating the relationship between injury severity and crash year.

Figure 5 shows that severe/fatal injuries are more common after 2020. The pandemic could be a leading reason why. Temporal effects on injury severity are intriguing, however, more analysis would be required to come to any concrete conclusions, in this regard.

4 Implications of our Findings

The primary finding of this analysis was that location plays a large role in the severity of pedestrian injury in traffic collisions. Such a finding suggests that those trying to prevent severe/fatal injuries in traffic collisions should focus on the locations identified by our models: crosswalks, bicycle lanes, and midblocks. Furthermore, our data suggests that severe/fatal injuries could have increased after the pandemic, and that one reason for this could be pedestrian actions after and during the pandemic period. These possibilities could be explored further in future projects.

Appendix
Data Dictionary

Field Name	Data Type	Description	Example	RANGE
PEDESTRIAN_Age.x	num	The age of the pedestrian.	15, 18, 89, 57	0 - 98
PEDESTRIAN_Gender	Factor	The gender of the pedestrian.	Female, Male,	3 levels
PEDESTRIAN_Condition.at.Time.of.Crash	Factor	The condition of the pedestrian at the time of the crash.	Normal, Asleep or Fatigued	7 levels
PEDESTRIAN_Actions.At.Time.Of.Crash	Factor	The actions of the pedestrian at the time of the crash.	Dart/Dash	13 levels
PEDESTRIAN_Actions.Prior.to.Crash	Factor	The actions of the pedestrian prior to the crash.	Moving Along Roadway	4 levels
PEDESTRIAN_Location.At.Time.Of.Crash	Factor	The location of the pedestrian at the time of the crash.	Bicycle facility	5 levels
PEDESTRIAN_Distracted	Factor	Whether or not the pedestrian was distracted.	Not Distracted	3 levels
CRASH_Year	Factor	The year the crash happened.	2016	9 levels
PERIOD	Factor	Whether or not the crash occurred before or after the pandemic.	Pandemic, Baseline	4 levels
CRASH_Month	Factor	The month the crash happened.	1, 2, 3	12 levels
CRASH_Day.of.	Factor	The day of the	Friday	7 levels

the.week		week the crash happened.		
CRASH_Route. Class	Factor	Description of the type of route on which the crash occurred.	Interstate	5 levels
CRASH_Light. Condition	Factor	Description of the light condition at the time of the crash.	Dark-Lighted	7 levels
CRASH_Road.S urface.Condition	Factor	Condition of the road surface at the time of the crash.	Dry	11 levels
CRASH_Crash. Specific.Locatio n	Factor	A detailed description of where the crash was located	Acceleration / Deceleration Lane	17 levels
CRASH_Type.o f.Intersection	Factor	The type of intersection where the crash occurred.	Five-point	8 levels
CRASH_Traffic way.Class	Factor	The class of trafficway where the accident occurred.	Non-trafficway	5 levels
CRASH_Highw ay.Type..FHWA	Factor	The type of highway where the accident occurred.	10 LANES DIVIDED (WIDTH 120'-143')	9 levels
VEHICLE_Roa dway.Grade	Factor	Description of the roadway where the accident occurred.	Downhill	6 levels
VEHICLE_Bod y.Type	Factor	Description of the type of body	(Sport) Utility Vehicle	18 levels

		of the vehicle.		
DRIVER_Gender.y	Factor	Gender of the driver.	Female	3 levels
DRIVER_Condition.at.Time.of.Crash.y	Factor	Condition of the driver at the time of the crash.	Apparently Normal	7 levels
DRIVER_Speeding.Related.y	Factor	Whether or not the driver was speeding.	Speeding	3 levels
DRIVER_Driver.Actions.1.y	Factor	Actions of the driver leading up to the crash.	Violated Traffic Rule	6 levels
DRIVER_Driver.Distracted.By.y	Factor	Whether or not the driver was distracted.	Distracted	3 levels
NETWORK_NumberOfThruLanes	Factor	Number of thru lanes at the site of the crash.	1, 2, 3...	11 levels
VIN_bodyclass	Factor	Class describing the vehicle body.	Cargo van	27 levels
VIN_vehicletype	Factor	Type of vehicle involved in the crash.	BUS	7 levels
PEDESTRIAN_Injury.Status	Factor	Status of the pedestrian after the crash.	Suspected Serious/Fatal Injury (A/K)	2 levels

Data Tidying

- **Variables Removed:** Sno., ProjectSiteId, CrashId, VehicleId, PEDESTRIAN_PersonId.x, NETWORK_MedianType, VEHICLE_Vehicle.Model.Year, VEHICLE_Total.Lanes.In.Roadway, VEHICLE_Roadway.Alignment, VEHICLE_Roadway.Alignment, VEHICLE_Traffic.Control.Device.Type, DRIVER_Age.y, NETWORK_SpeedChangeLane, NETWORK_LaneWidth, 'Front bumper height', PEDESTRIAN_Injury_Severity_KA2.BC1.O0
- **Summary of Changes made:**
 - PEDESTRIAN_Condition.at.Time.of.Crash: Null observations were put under the “Unknown” label along with observations labeled “Other” and “Not Applicable”.

The “Apparently Normal” category was changed to just “Normal”. The category "Under the Influence of Medications/Drugs/Alcohol" was renamed to intoxicated. And the category "Emotional (depressed, angry, disturbed, etc.)" was renamed to emotional.

- PEDESTRIAN_Actions.[Prior.to](#).Crash: Null observations were put under the “Unknown” label along with “Other”, “None”, and "In Roadway - Other" = "Unknown" observations. The "Crossing Roadway" and "Waiting to Cross Roadway" categories were collapsed to “Crossing”. The "Walking/Cycling along Roadway Against Traffic (In or Adjacent to Travel Lane)", "Walking/Cycling along Roadway Along Traffic (In or Adjacent to Travel Lane)", "Walking/Cycling on Sidewalk" and "Adjacent to Roadway (e.g., Shoulder, Median)" categories were collapsed to “Moving Along Roadway”. The "Working in Trafficway for Incident Response" was renamed “Working in Trafficway”
- PEDESTRIAN_Actions.[At.Time.Of](#).Crash: Null observations were put under the "Unknown" label along with “Other”, "Not Visible (Dark Clothing, No Lighting, etc.)", and "Not Applicable".
- PEDESTRIAN_Location.[At.Time.Of](#).Crash: : Null observations were put under the “Unknown” label along with “Other”. The "Intersection-Marked Crosswalk", "Intersection-Unmarked Crosswalk", and "Intersection – Other" categories were collapsed to “Intersection”. The "Travel Lane-Other Location", "Midblock-Marked Crosswalk", "Shoulder/Roadside", and "Median/Crossing Island" categories were collapsed to “Midblock / Travelway”. The "Sidewalk", "Shared-Use Path or Trail", "Non-Trafficway Area", and "Driveway Access" categories were collapsed to “Off-road / Non-traffic facility”. The "Bicycle Lane" and "Sharrows" categories were collapsed to “Bicycle facility”.
- PEDESTRIAN_Distracted: All null observations were put under “unknown if distracted”. All categories were then collapsed into Not Distracted, Distracted, and Unknown if Distracted.
- CRASH_Route.Class: Null observations were put under the “Unknown” label.
- CRASH_Light.Condition: The “Other” category was added to “Unknown”.
- CRASH_Road.Surface.Condition: The “Other” category was added to “Unknown”.
- CRASH_CRASH.Specific.Location: The “Other” category was added to “Unknown”.
- CRASH_Highway.Type..FHWA: Null observations were put under the “Unknown” label.
- Vehicle_Roadway.Grade: Null observations were put under the “Unknown” label.
- Vehicle_Body.Type: Null observations were put under the “Unknown” label along with “Other” observations.
- DRIVER_Gender.y: Null observations were put under the “Unknown” label.

- DRIVER_Condition.[at.Time.of](#).Crash.y = Null observations were put under the “Unknown” label along with observations labeled “Other” and “Not Applicable”.
- DRIVER_Speeding.Related.y: Null observations were put under the “Unknown” label. "Exceeded Speed Limit", “Racing”, and "Too Fast for Conditions" were collapsed to “Speeding”. “No” was renamed to “Not Speeding”.
- Driver_Driver.Actions.1.y = Null observations were put under the “Unknown” label along with observations labeled “Other” and “Not Applicable”. The "Disregarded Other Road Markings", "Disregarded Other Traffic Sign", "Failed to Keep in Proper Lane", "Failed to Yield Right-of-Way", "Ran Red Light", "Ran Stop Sign", and "Wrong Side or Wrong Way" categories were collapsed to “Violated Traffic Rules”. The "Followed Too Closely", "Improper Backing", "Improper Passing", "Improper Turn", "Over-Correcting/Over-Steering", "Swerved or Avoided Due to Wind, Slippery Surface, Motor Vehicle, Object, Non-Motorist in Roadway, etc.", and "Ran Off Roadway" were collapsed to “Driving Error”. The "Operated Motor Vehicle in Inattentive, Careless, Negligent, or Erratic Manner", and "Operated Motor Vehicle in Reckless or Aggressive Manner" were collapsed to “Negligent Driving”.
- DRIVER_Driver.[Distracted.By](#).y: The "Manually Operating an Electronic Communication Device (texting, typing, dialing)", "Other Activity, Electronic Device", "Outside the Vehicle (includes unspecified external distractions)", "Passenger", "Talking on Hand-Held Electronic Device", "Talking on Hands-Free Electronic Device", and "Other Inside the Vehicle (eating, personal hygiene, etc.)" categories were collapsed to “Distracted”. Null observations were added to the “unknown if distracted category”.
- NETWORK_NumberOfThruLanes: Null observations were put under the “Unknown” label.
- VIN_bodyclass: Null observations were put under the “Unknown” label.
- PEDESTRIAN_Injury.Status: All observations corresponding to the "Suspected Minor Injury (B)" and "Possible Injury (C)" were filtered out. The categories "Fatal Injury (K)" and "Suspected Serious Injury (A)" were collapsed to "Suspected Serious/Fatal Injury (A/K)".